

Khulna University of Engineering & Technology

Department of Electronics and Communication Engineering

Course No: ECE 2200

Course Title: Electronic Circuits Design Laboratory

Open-Ended Project Report on

Wireless Power Transfer

– Short-Range Inductive Mobile Charging System

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1. Introduction

Wireless power transfer (WPT) enables electrical energy to be transferred from a power source to a load without a direct conductive connection. Among its different techniques, inductive power transfer is widely used for short-range applications due to its simple operating principle and practical implementation.

This project, entitled “Wireless Power Transfer: Short-Range Inductive Mobile Charging System,” focuses on designing and implementing a laboratory-scale system using electromagnetic induction. The transmitter generates an alternating magnetic field through a coil, inducing a voltage in a nearby receiver coil.

The transmitter uses a 12 V DC supply, a KA3525A PWM controller, and two IRFZ44N MOSFETs to drive a center-tapped coil. The receiver consists of an 18–20-turn coil, a compensation capacitor, a Schottky-diode bridge rectifier, filter capacitors, and a 7805 voltage regulator. The regulated output is connected to a USB Type-C breakout interface for low-current charging demonstrations.

The system targets an operating frequency of approximately 70–80 kHz and an output close to 5 V under suitable coil alignment and loading conditions. The project also examines how coil spacing, alignment, orientation, switching frequency, and load affect the received voltage and power-transfer efficiency.

The primary objective is to demonstrate electromagnetic induction and the practical integration of switching circuits, magnetic coupling, rectification, filtering, and voltage regulation.

2. Objectives

- To design and build a simple wireless power transfer system using transmitter and receiver coils to transfer electrical power through air without direct wire connections
- To study the principles of electromagnetic induction and calculate the required circuit parameters and component values for the transmitter and receiver circuits
- To measure output voltage, current, and efficiency, and analyze the effects of coil spacing, alignment, orientation, frequency, and load on system performance

3. Theory

3.1 Principle of Wireless Power Transfer

Wireless power transfer in this project is based on electromagnetic induction. When alternating current flows through the transmitter coil, it produces a time-varying magnetic field. Some of this magnetic flux passes through the nearby receiver coil, inducing an electromotive force (EMF) in it according to Faraday's law of electromagnetic induction.

Faraday's law is expressed as:

$$\varepsilon = -N \times \frac{d\Phi}{dt}$$

Where:

- ε = induced electromotive force (V)
- N = number of turns in the coil
- Φ = magnetic flux (Wb)
- $d\Phi/dt$ = rate of change of magnetic flux

The negative sign represents Lenz's law, which states that the induced EMF opposes the change in magnetic flux responsible for producing it.

The induced voltage depends on the magnetic field strength, number of turns, coil geometry, operating frequency, relative coil position, and magnetic coupling between the coils.

3.2 Electromagnetic Induction and Mutual Inductance

The transmitter and receiver coils are magnetically coupled. The changing current in the transmitter produces a changing magnetic field that links with the receiver coil. Mutual inductance describes the strength of this magnetic interaction.

The induced voltage can be expressed approximately as:

$$V_r = M \times \frac{dI_t}{dt}$$

Where:

- V_r = induced voltage in the receiver
- M = mutual inductance between the coils
- I_t = transmitter-coil current
- $\frac{dI_t}{dt}$ = rate of change of the current in the transmitter

Higher mutual inductance generally produces a higher induced voltage for the same rate of change of transmitter current. Mutual inductance is affected by coil spacing, alignment, orientation, and coil dimensions.

3.3 Resonance in the Coil Circuit

Resonance can improve energy transfer by compensating for the inductive reactance of a coil using a capacitor. The theoretical resonant frequency of an ideal LC circuit is:

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Where:

- f_r = resonant frequency (Hz)
- L = inductance (H)
- C = capacitance (F)

For the transmitter coil, the inductance approximately 14.4 µH and a parallel compensation capacitor of 300 nF.

Therefore,

$f_r \approx 76.5$ kHz

For the receiver coil, the inductance approximately 40.5 µH and a parallel compensation capacitor of 100 nF.

Therefore,

$f_r \approx 79.1$ kHz

This theoretical value is close to the proposed operating range. However, the actual resonant frequency may differ because of component tolerances, coil resistance, parasitic capacitance, loading, and interaction between the transmitter and receiver. The final operating frequency should therefore be verified experimentally.

3.4 Transmitter Switching Circuit

The transmitter uses a KA3525A PWM controller to generate alternating control signals for two IRFZ44N MOSFETs. The MOSFETs switch current through the two halves of the center-tapped transmitter coil, generating a high-frequency alternating magnetic field.

The proposal specifies a 12 V DC supply, a center-tapped 6-0-6-turn transmitter coil, and an initial switching frequency near 75 kHz. The controller's timing components determine its

oscillator frequency, while its alternating outputs operate at approximately half the oscillator frequency in the proposed configuration.

Gate resistors help control switching transitions, pull-down resistors prevent unintended MOSFET turn-on, and zener protection and snubber networks help reduce switching-related electrical stress.

3.5 Receiver Rectification and Filtering

The receiver coil generates an AC voltage due to the changing magnetic field. Since the intended output is DC, the induced voltage passes through a full-wave bridge rectifier consisting of four 1n5819 Schottky diodes.

Schottky diodes are suitable for high-frequency rectification because they generally have a relatively low forward voltage drop and fast switching characteristics.

A filter capacitor connected after the bridge rectifier reduces the ripple in the rectified voltage. A simplified approximation for the peak-to-peak ripple voltage is:

$$V_r(pp) = \frac{I}{f_{ripple} \times C}$$

Where:

- $V_r(pp)$ = peak-to-peak ripple voltage (V)
- I = load current (A)
- f_{ripple} = rectifier ripple frequency (Hz)
- C = filter capacitance (F)

For a full-wave rectifier, the ripple frequency is twice the AC input frequency. This approximation assumes relatively small ripple and does not account for every diode, source-impedance, and capacitor loss.

3.6 Voltage Regulation

The receiver uses a 7805 linear voltage regulator to obtain an approximately 5 V DC output. The regulator helps maintain a stable output when the input voltage and load vary within its operating limits.

The regulator requires sufficient input voltage above its output voltage. If the rectified and filtered voltage falls below the regulator's required input level, the output may drop below 5 V.

The power dissipated by a linear regulator is approximately:

$$P_D = (V_{in} - V_{out}) \times I_{out}$$

Where:

- P_D = regulator power dissipation (W)
- V_{in} = regulator input voltage (V)
- V_{out} = regulated output voltage (V)
- I_{out} = output current (A)

Excessive power dissipation can increase the regulator's temperature and reduce performance. Therefore, input voltage, output current, and thermal conditions must be considered during testing.

4. Motivation & Background

The growing use of smartphones, portable electronics, and battery-powered devices has increased interest in convenient methods of delivering electrical power. Conventional charging generally requires a physical connection between the charger and the device. Wireless power transfer offers an alternative by allowing electrical energy to cross a short air gap through magnetic coupling.

The motivation behind this project is to understand the engineering principles used in practical wireless charging systems and to apply theoretical knowledge to a working electronic circuit. The project integrates several important topics from electronic circuit design, including PWM control, MOSFET switching, inductance, resonance, electromagnetic induction, rectification, filtering, and voltage regulation.

The project also addresses practical design challenges. Energy transfer decreases when the coils are poorly aligned or separated by a larger distance. Changes in operating frequency and load can affect the received voltage, output current, component temperature, and overall efficiency. Consequently, successful implementation requires more than assembling the circuit; it requires measurement, troubleshooting, and optimization.

The prototype is intended for educational demonstration and low-power applications. It is not designed or certified as a commercial fast charger. The project provides a foundation for understanding more advanced wireless power systems and their engineering limitations.

5. Required Apparatus

5.1 table-1: table for electronic components

Sl. No.	Component	Specification or rating	Quantity
1	DC power supply	12 V DC	1
2	KA3525A PWM controller	DIP-16	1
3	IRFZ44N MOSFET	55 V class	2
4	Transmitter coil	Center-tapped 6-0-6 turns, approximately 5 cm diameter	1
5	Receiver coil	Approximately 18–20 turns, approximately 5 cm diameter	1
6	Enameled copper wire	24 AWG for transmitter coil	As required
7	1N5819 Schottky diode	3 A, 40 V class	4
8	7805 voltage regulator	5 V fixed output	1
9	Polypropylene film capacitors	Compensation and switching circuits	As required
10	Electrolytic capacitors	470–1000 µF and 10 µF	As required
11	Ceramic capacitors	100 nF and other required values	As required
12	Resistors	Gate, timing, pull-down, and snubber networks	As required
13	Zener diodes	Gate protection	As required
14	Fuse and current-limiting components	Input protection	As required
15	USB Type-C breakout and cable	5 V output interface	1 set
16	Breadboard or PCB, connectors, and mounting materials	Circuit assembly	As required
17	Heat sink	For 7805 regulator, if required	1

5.2 Testing and Measurement Equipment

1. Digital multimeter for measuring DC voltage, current, and resistance.
2. Oscilloscope for observing switching waveforms and receiver signals.
3. Soldering iron and soldering accessories for PCB assembly.
4. Connecting wires, probes, and test leads.
5. Electronic load or suitable power resistors for controlled load testing.
6. Computer with Proteus or equivalent circuit simulation software, if used.
7. Breadboard for the initial circuit implementation.
8. PCB fabrication and assembly tools, as required.

These instruments support circuit verification, troubleshooting, measurement of electrical parameters, and evaluation of the system's performance.

6. Circuit Diagram & Analysis (Experimental Setup)

6.1 Block Diagram of the System

The complete wireless power transfer system consists of two main sections: the transmitter and the receiver.

Transmitter section:

12 V DC Supply → Input Protection → KA3525A PWM Controller → IRFZ44N MOSFET Switching Stage → Transmitter Coil

Wireless coupling:

Transmitter Coil → Alternating Magnetic Field Across Air Gap → Receiver Coil

Receiver section:

Receiver Coil → Compensation Capacitor → Schottky Bridge Rectifier → Filter Capacitors → 7805 Voltage Regulator → USB Type-C Output

The compensation capacitor and coil form a resonant network. The actual circuit connections should follow the final circuit diagram used to assemble the prototype.

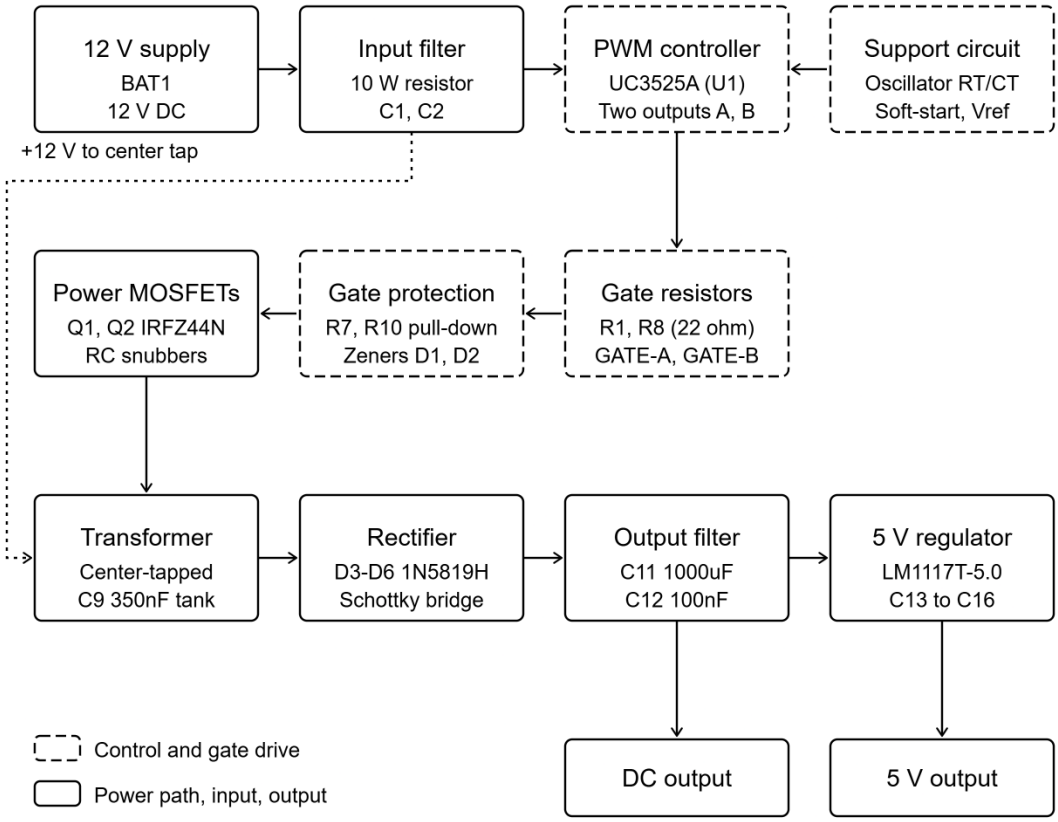


Figure 1: Block diagram of the wireless power transfer system.

6.2 Transmitter Circuit Design

The transmitter circuit generates the alternating magnetic field required for wireless energy transfer. It consists of a 12 V DC supply, an input protection network, a KA3525A PWM controller, two IRFZ44N MOSFETs, a center-tapped transmitter coil, and associated timing and protection components.

The KA3525A generates alternating control signals for the two MOSFETs. The timing network is designed to provide an operating frequency close to the proposed 75 kHz switching range. The two MOSFETs alternately conduct through the corresponding halves of the center-tapped coil, producing an alternating magnetic field around the transmitter.

The transmitter coil is constructed using 24 AWG enameled copper wire, with a center-tapped 6-0-6-turn arrangement and an approximate diameter of 5 cm.

The transmitter components serve the following purposes:

- KA3525A controller: Generates the switching control signals.
- IRFZ44N MOSFETs: Act as electronic switches for driving the transmitter coil.
- Gate resistors: Limit gate-charging current and control switching transitions.

- 10 kΩ pull-down resistors: Keep the MOSFET gates at a defined low voltage when the drive signal is inactive.
- Zener diodes: Provide gate-voltage protection when correctly selected and connected.
- Snubber networks: Help limit switching transients and reduce voltage stress.
- Input protection: Helps protect the circuit against excessive current and supply-related faults.
- Polypropylene capacitor: Forms part of the transmitter's compensation or switching network as specified in the circuit design.

The proposed timing components are R_t = 10 kΩ, C_t = 1 nF, and a 100 Ω discharge/dead-time resistor. The proposal estimates an oscillator frequency near 150 kHz and alternating output signals near 75 kHz. These are design estimates and should be confirmed by measuring the actual gate-drive waveforms.

6.3 Receiver Circuit Design

The receiver circuit captures the alternating magnetic energy produced by the transmitter and converts it into usable DC power.

The receiver coil contains approximately 18–20 turns and has an approximate diameter of 5 cm. The proposal specifies a measured coil inductance of approximately 40.5 µH, together with a 100 nF polypropylene compensation capacitor.

The receiver operates as follows:

1. Energy reception: The alternating magnetic field induces an AC voltage in the receiver coil.
2. Resonant compensation: The capacitor compensates for the coil's inductive reactance near the intended operating frequency.
3. Rectification: Four 1n5819 Schottky diodes form a bridge rectifier to convert the received AC into pulsating DC.
4. Filtering: A 1000 µF capacitor and a 100 nF bypass capacitor reduce voltage ripple.
5. Voltage regulation: A 7805 regulator produces a nominal 5 V DC output when its input voltage and thermal conditions are suitable.
6. Output connection: The regulated output is connected to the USB Type-C breakout for testing with an appropriate electronic load.

The proposal also specifies 10 µF and 100 nF capacitors around the regulator for supply bypassing and stability, according to the selected circuit arrangement and regulator requirements.

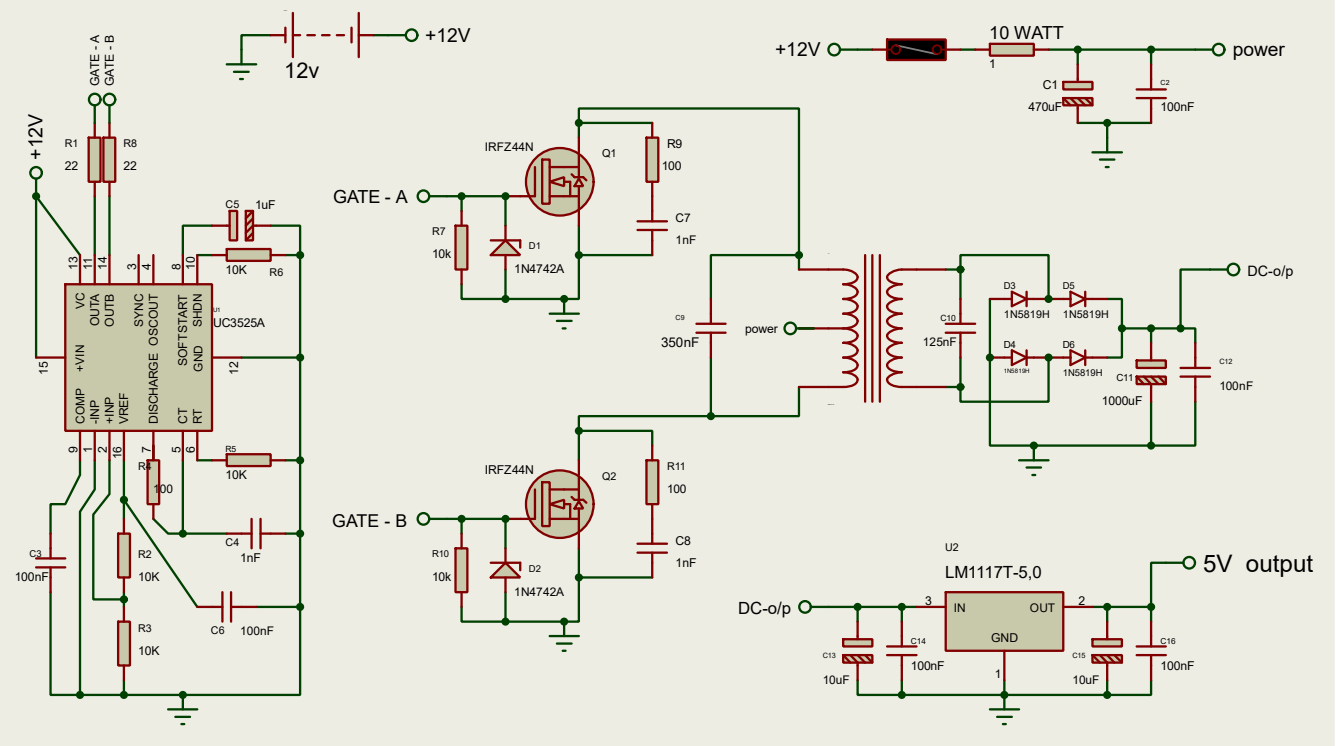


Figure 2: circuit diagram of the wireless power transfer circuit.

6.5 Experimental Procedure

The following procedure is used to evaluate the assembled prototype.

1. Inspect the transmitter and receiver circuits to verify component polarity, coil connections, grounding, and supply connections.
2. Apply the 12 V DC supply and check whether the KA3525A generates the expected alternating control signals.
3. Observe the MOSFET gate waveforms with an oscilloscope and verify that the switching frequency and timing are suitable.
4. Place the receiver coil close to the transmitter coil, initially using the proposed separation of approximately 2–5 mm.
5. Measure the induced receiver voltage before and after rectification, where practical.
6. Measure the filtered DC voltage at the receiver and check whether it provides sufficient input for the 7805 regulator.
7. Measure the regulated output voltage with no load and with suitable test loads.
8. Measure the transmitter's DC input voltage and current and the receiver's output voltage and current.

9. Repeat the measurements while changing coil separation, alignment, orientation, operating frequency, and load, if these tests were performed.
10. Record the measurements, calculate output power and efficiency, and identify the conditions that provide the most stable output.

Testing should begin with a current-limited supply where possible. The MOSFETs, coils, rectifier, and voltage regulator should be monitored for excessive heating. The USB Type-C interface should not be assumed to provide complete USB-C charging compatibility merely because a regulated 5 V supply is connected to its VBUS pin.

7. Results and Discussion

7.1 Functional Verification

- The transmitter and receiver circuits were assembled and tested
- Wireless power transfer was successfully demonstrated across an approximate distance of 2.5 cm (1 inch)
- The receiver circuit used a Schottky diode bridge, filter capacitors, and a 7805 voltage regulator
- The circuit was first tested on a breadboard and later assembled on a PCB

7.2 Discussion of Results

- The prototype demonstrated short-range wireless power transfer through electromagnetic induction
- The received power depends on coil coupling, alignment, operating frequency, and load conditions
- Further measurements are required to determine the actual output power, voltage stability, and transfer efficiency

7.3 Limitations and Possible Improvements

- **Limited transfer distance**: Optimize coil design and positioning to improve magnetic coupling
- **Coil alignment**: Improve alignment to maintain effective power transfer
- **Power losses**: Reduce switching, coil resistance, and rectification losses

- **Voltage regulation**: Improve the 7805 regulator's thermal performance or consider a switching regulator
- **Limited testing**: Measure input/output power, efficiency, output stability, and component temperature
- **Future development**: Investigate maximum transfer distance and performance under different loads

8. Socio-Economic Impact and Sustainability

8.1 Socio-Economic Impact

Wireless power transfer has potential applications in portable electronics, medical equipment, embedded sensors, and industrial instrumentation. By eliminating the need for a direct conductive connection during energy transfer, the technology can improve convenience and reduce wear on physical charging connectors in suitable applications.

The project develops practical skills in electronic circuit design, component selection, circuit assembly, measurement, troubleshooting, and technical reporting. These skills are relevant to electronics engineering, embedded systems, power electronics, and wireless technology development.

The prototype also demonstrates that fundamental wireless charging concepts can be explored using commonly available electronic components and laboratory equipment. This makes it suitable for educational demonstrations and introductory engineering projects.

However, the prototype is a low-power laboratory system and should not be considered a replacement for commercial charging systems without further engineering, safety testing, and compliance verification.

8.2 Environmental Sustainability

The system uses electrical energy to demonstrate wireless transfer over a short distance. Its environmental performance depends on its overall efficiency, component lifetime, operating conditions, and the source of its input electricity.

The following practices can improve sustainability:

- Use components efficiently and avoid unnecessary power dissipation.
- Switch off the DC supply when the prototype is not in use.
- Reuse functional electronic components in future laboratory projects.
- Handle solder, electronic waste, and damaged electronic components responsibly.
- Minimize unnecessary material consumption during circuit construction.

- Improve the system's efficiency before considering larger-scale applications.

A key limitation is that wireless transfer introduces losses in the transmitter, coils, receiver, and regulator. Therefore, wireless charging is not automatically more energy-efficient than wired charging. Measuring and reducing these losses is important for evaluating its environmental benefits.

8.3 Safety Considerations

The prototype should be operated using the specified low-voltage DC supply. The circuit should be disconnected before modifying wiring or replacing components. The MOSFETs, coils, diodes, and voltage regulator should be checked for excessive heating during operation.

The output should be verified before connecting any electronic device. Direct connection to a mobile phone or other valuable equipment should be avoided until output stability and interface compatibility have been confirmed.

9. Project Planning

The project was completed through a sequence of research, theoretical design, component selection, circuit construction, testing, PCB implementation, and documentation. The initial stages focused on understanding the operating principles and preparing the circuit. The later stages involved implementing the circuit on a breadboard, transferring the design to a PCB, and preparing the final report.

The actual project schedule is summarized below.

Week	**Activities performed**	**Outcome**
First week	Studied wireless power transfer principles, reviewed reference circuits, calculated component values, and finalized the initial circuit design.	Theoretical design and component requirements established.
Second week	Procured the required components and constructed the transmitter and receiver coils.	Components collected and coils prepared.
Third week	Assembled the KA3525A control circuit and IRFZ44N MOSFET switching stage, then checked the transmitter operation.	Transmitter circuit assembled and tested.
Fourth week	Assembled the receiver coil, compensation capacitor, rectifier, filter, and voltage regulator stages.	Receiver circuit assembled and integrated with the transmitter.

Fifth week	Implemented the complete circuit on a breadboard and performed initial functional testing and troubleshooting.	Breadboard prototype completed and its operation evaluated.
Sixth week	Designed and fabricated the PCB, transferred the circuit to the PCB, and assembled the components.	PCB implementation of the wireless power transfer circuit completed.

10. Budget Analysis

The following budget is based on the estimated local-market component costs provided in the project proposal.

Sl. No.	**Component**	**Quantity**	**Estimated cost (BDT)**
1	KA3525A PWM controller	1	35
2	IRFZ44N MOSFETs	2	120
3	Transmitter and receiver coils	Length = 5.5 meter	300
4	Polypropylene film capacitors	4	40
5	1n5819 Schottky diodes	4	10
6	7805 regulator	1	15
7	Electrolytic and ceramic capacitors	15	75
8	Resistors	11	10
9	Zeners (15V)	2	10
10	Fuse (3A)	1	5
11	Current-limiting resistor	1	10
12	USB Type-C cable	1	80
13	Heat Sink	3	30
14	Miscellaneous materials	As required	100
Total estimated cost (BDT)			840

The estimated component cost is BDT 840, based on the initial proposal. The largest expenses are the coils, capacitors, and protection components. The actual cost may vary depending on market prices, PCB fabrication, and additional materials.

11. Conclusion

The project successfully demonstrated short-range wireless power transfer over an approximate distance of 2.5 cm (1 inch) using electromagnetic induction. The system integrates PWM control, MOSFET switching, a receiver coil, rectification, filtering, and voltage regulation.

The receiver coil's theoretical resonant frequency is approximately 79.1 kHz, based on an inductance of 40.5 µH and a 100 nF capacitor. The project provided practical experience in circuit design, component selection, breadboard testing, and PCB implementation.

The project also helped develop an understanding of the factors affecting wireless power transfer, including coil alignment, separation, and circuit losses. The use of readily available electronic components makes the prototype suitable for educational and laboratory applications.

With an estimated component cost of BDT 840, the prototype offers an affordable demonstration of wireless energy transfer. However, further measurements of output voltage, current, efficiency, and component temperature are needed to evaluate its performance more accurately.

Future improvements could focus on increasing transfer efficiency, improving output stability, and optimizing coil positioning. Replacing the linear voltage regulator with a suitable switching regulator may also reduce power losses. Overall, this project provides a practical foundation for understanding inductive wireless charging and developing more efficient wireless power transfer systems.